Chapter 1

Intro to Galois Theory

1.1 Meeting #1 Notes

These and subsequent notes are based on [AM98].

1.1.1 Normal Extensions

Recall the notation from the book:

- $G := \mathrm{Aut}(L/K) \triangleq \{\sigma \in \mathrm{Aut}(L) : \forall_{x \in K}\ \sigma(x) = x\}$.
- $\mathrm{Fix}(L/K) = \{x \in L : \forall_{\sigma \in G}\ \sigma(x) = x\}$.

Clearly, $K \subseteq \mathrm{Fix}(L/K)$, but the converse is not always the case. When it is, though, the extension L/K is called **normal**.

Definition 1.1. *The following are equivalent definitions of L/K being a **normal extension**:*

(a) $[L : K]$ is finite and $Fix(L/K) = K$.

(b) any irreducible polynomial $f \in K[T]$ such that f has at least one root in L splits in L.

Example. *We demonstrate the definition (a) on two examples:*

(a) $L := \mathbb{Q}(\sqrt{2})$ is a normal extension of $\mathbb{Q}$. Indeed, every $\mathbb{Q}$-automorphism of L is determined where it sends $\sqrt{2}$. The only posibilities are $id : \sqrt{2} \mapsto \sqrt{2}$ and $\sigma : \sqrt{2} \mapsto -\sqrt{2}$. To find fixed field, we notice that $a + b\sqrt{2} \in Fix(L/K)$ iff $a + b\sqrt{2} = a - b\sqrt{2}$ which implies $b = 0$. Thus, $L/\mathbb{Q}$ is a normal extension.

(b) $L := \mathbb{Q}(\sqrt[3]{2})$ is not a normal extension. Similarly, automorphisms are determined where they send roots of $T^3 - 2$, which are $\sqrt[3]{2}$, $\sqrt[3]{2}\omega$ and $\sqrt[3]{2}\omega^2$, but the latter two roots are not in $\mathbb{Q}(\sqrt[3]{2})$. Thus, $G = \{id\}$ and so $Fix(L/\mathbb{Q}) = L$.

In general, to fix the issue in (b), it suffices to take the extension $L := \mathbb{Q}(\sqrt[p]{2}, \zeta_p)$ where ζ_p is the principle root of 1.

Theorem 1.1. *If L/K is a normal extension, then $[L : K] = \#Aut(L/K)$.*

Proof. Let $G = \{\sigma_1, \ldots, \sigma_n\}$. Since this is a group, the identity occurs, say, $\sigma_1 := \mathrm{id}$. Suppose $[L : K] > n$. Then, there exist $\alpha_1, \ldots, \alpha_{n+1} \in L$ which are linearly independent

over K. Thus, there exists a non-trivial solution in L to the system of equations

$$\sum_{j=1}^{n+1} x_j \sigma_i(\alpha_j) = 0, \quad 1 \leq i \leq n.$$

Note that non-trivial solutions do not lie in K: otherwise, since σ_1 is the identity, the first equation would be the dependence between $\alpha_1, \ldots, \alpha_{n+1}$.

Among all non-trivial solutions $\mathbf{x}_1, \ldots, \mathbf{x}_{n+1}$ we choose one which has the least number of elements different from 0: let, for instance, this solution be $(z_1, \ldots, z_r, 0, \ldots, 0)$ where all $z_i \neq 0$. Moreover, $r \neq 1$, because this would imply $z_1 = 0$. Also without loss of generality let $z_r = 1$. Thus,

$$z_1 \sigma_i(\alpha_1) + \cdots + z_{r-1} \sigma_i(\alpha_{r-1}) + \sigma_i(\alpha_r) = 0, \quad 1 \leq i \leq n.$$

Since the whole solution cannot lie in L, there is some element in $L \setminus K$, say, $z_1 \in L \setminus K$. There is some automorphism σ_k such that $\sigma_k(z_1) \notin K$. Note that $\sigma_k\sigma_1, \ldots, \sigma_k\sigma_n$ is a permutation of $\sigma_1, \ldots, \sigma_n$, thus, applying σ_k to equation above,

$$\sigma_k(z_1) \cdot \sigma_k\sigma_j(\alpha_1) + \cdots + \sigma_k(z_{r-1}) \cdot \sigma_k\sigma_j(\alpha_{r-1}) + \sigma_k\sigma_j(\alpha_r) = 0.$$

And thus by renaming, we get:

$$\sigma_k(z_1) \cdot \sigma_i(\alpha_1) + \cdots + \sigma_k(z_{r-1}) \cdot \sigma_i(\alpha_{r-1}) + \sigma_i(\alpha_r) = 0.$$

Now, subtracting this equation from the previous,

$$(z_1 - \sigma_k(z_1))\sigma_i(\alpha_1) + \cdots + (z_{r-1} - \sigma_k(z_{r-1}))\sigma_i(\alpha_{r-1}) = 0.$$

which is a non-trivial solution to the system having fewer than r elements, contrary to the choice of r.

Theorem 1.2. *L/K is a normal extension iff L is the splitting field of a separable $f \in K[T]$.*

Proof. Suppose L splits f. If all roots of f are in K, then the proposition is trivial, since thus $L = K$ and only the unit automorphism leaves K fixed.

Thus suppose f has $n > 1$ roots in $L \setminus K$. We proceed by induction: suppose our assumption holds for all pairs of fields with fewer than n roots of f outside K. Let $f(T) = f_1(T) \ldots f_r(T)$ be a factorization of f into irreducible factors. Assume that one of those has a degree greater than one (otherwise f splits in K). Suppose $\deg f_1 = s > 1$ and let $\alpha_1 \in L$ be the root of f_1. Then, $[K(\alpha_1) : K] = s$. By inductive assumption, $L/K(\alpha_1)$ is a normal extension and thus each element of $L \setminus K(\alpha_1)$ is moved by some automorphism $\sigma_1 : L \to L$ that leaves $K(\alpha_1)$ fixed.

As f is separable, the roots $\alpha_1, \ldots, \alpha_s$ of f_1 are distinct elements of L. Thus, there exist isomorphisms $\sigma_i : K(\alpha_1) \to K(\alpha_i)$ leaving K fixed and $\alpha_i = \sigma_i(\alpha_1)$. By applying one of the Theorems before, as L is a splitting field of f in $K(\alpha_1)$, it is also a splitting field in $K(\alpha_i)$. Thus, we can extend each isomorphism to an automorphism $\sigma_i : L \to L$ that leave K fixed and mapping $\alpha_1 \mapsto \alpha_i$.

Now suppose θ is fixed under all automorphisms of L that leave K fixed. In particular, it is fixed by all automorphism fixing $K(\alpha_1)$, and thus it lies in $K(\alpha_1)$, so $\theta = c_0 +$

$c_1\alpha_1 + \cdots + c_{s-1}\alpha_1^{s-1}$. By appling each σ_i and noting that $\sigma_i(\theta) = \theta$, we obtain $\theta = c_0 + c_1\alpha_i + \cdots + c_{s-1}\alpha_i^{s-1}$. Thus, the polynomial $(c_0 - \theta) + c_1T + \cdots + c_{s-1}T^{s-1}$ has s roots $\alpha_1, \ldots, \alpha_s$. But it has a smaller degree than s, thus it is a zero polynomial, implying $c_0 = \theta$. This shows that $\theta \in K$.

We next first prove the following lemma.

Lemma 1.1. *If L/K is a normal extension, then L is a separable extension of K. Moreover, any element of L is a root of an equation over K which splits completely in L.*

Proof of lemma. Let $\{\sigma_1, \ldots, \sigma_s\} = \mathrm{Aut}(L/K)$. Let $\alpha \in L$ and let $\alpha_i = \sigma_i(\alpha)$. Note that $\sigma_j(\alpha_i) = \sigma_j(\sigma_i(\alpha)) = \sigma_j\sigma_i(\alpha) = \sigma_k(\alpha) = \alpha_k$ for some k. Thus, each σ_i permutes $(\alpha_1, \ldots, \alpha_s)$. Now consider the polynomial $f(T) := \prod_{i=1}^{s}(T - \alpha_i)$. Clearly, σ_i fix coefficients of f, and since L/K is normal, $f \in K[T]$. If $g \in K[T]$ also has α as a root, then applying automorphism to $g(\alpha) = 0$, we obtain $g(\alpha_i) = 0$, so $\deg g > s$. Hence f is irreducible, and the lemma is established.

Back to the theorem. Now assume that L/K is a normal extension. Let $\omega_1, \ldots, \omega_s$ be the basis of L/K. Let f_i be the separable polynomial having ω_i as a root. Then, L is a splitting field of $f(T) = \prod_{j=1}^{t} f_j(T)$.

Definition 1.2. *Let $f \in K[T]$ and L the splitting field of f. Then, $Aut(L/K)$ is the **group of the equation** $f(T) = 0$.*

Theorem 1.3 (Fundamental Theorem)**.** *If $f \in K[T]$ is separable and G is the group of equation $f(T) = 0$ with L the splitting field of f, then:*

(a) There is a bijection between subgroups $H \leq G$ and $K \subseteq F \subseteq L$ given by $H \mapsto \{x \in L : \forall_{\sigma \in H}\sigma(x) = x\}$, $K \mapsto Gal(E/K)$.

(b) Intermediate field F is normal over K iff $Gal(L/F) \triangleleft G$. In fact, $Gal(F/K) \cong G/Gal(L/F)$.

(c) For any intermediate F, $[L : F] = |Gal(L/F)|$ and $[F : K] = [G : Gal(L/F)]$ and $[E : F] = |G|$.

1.1.2 Finite Fields

Nothing new in particular. If K is a finite field, $K^\times$ is cyclic.

Theorem 1.4. *Two finite fields with the same number of elements are isomorphic.*

Proof. Assume K and K' are of order $q = p^n$. Then, $K/\mathbb{F}_p$ and $K'/\mathbb{F}_p$ are splitting fields of $T^{q-1} - 1$. By Theorem 10, the isomorphism between two base fields $\mathbb{F}_p$ can be extended to the isomorphism between K and K'.

1.1.3 Roots of Unity

Suppose K is a field with characteristic p and L is the splitting field of the polynomial $T^n - 1$ where p does not divide n. Then, L is a field generated out of K by the adjunction of a primitive n^{th} root of unity. Since derivative of $T^n - 1$ is nT^{n-1} has only the root 0, L/K is a normal extension. If $1, \zeta_n, \zeta_n^2, \ldots, \zeta_n^{n-1}$ are the roots of $T^n - 1$ in L, they form a

cyclic group under multiplication.

Theorem 1.5. *If L is generated from K by a primitive n^{th} root of unity, then the group $G := Aut(L/K)$ is abelian for any n and cyclic if n is the prime number.*

Clearly $L = K(\zeta_n)$. Thus, if σ and τ are distinct elements of G, $\sigma(\zeta_n) \neq \tau(\zeta_n)$. But $\sigma(\zeta_n) = \zeta_n^{n_\sigma}$. Moreover, $\tau\sigma(\zeta_n) = \tau(\zeta_n^{n_\sigma}) = \tau(\zeta_n)^{n_\sigma} = \zeta_n^{n_\sigma n_\tau} = \sigma\tau(\zeta_n)$. Thus,

$$\mathrm{Aut}(K(\zeta_n)/K) \to (\mathbb{Z}/n\mathbb{Z})^\times, \quad \sigma(\zeta_n) = \zeta_n^{n_\sigma} \mapsto n_\sigma$$

is a homomorphism. This is also clearly an isomorphism. If n is prime, this group forms a cyclic group.

1.2 Meeting #2 Notes

Comment: This is a better and more structured way to prove theorem from the previous meeting.
Recall. We say that L/K is **normal** if $\mathrm{Fix}(L/K) = K$ and $[L : K] < \infty$.

Lemma 1.2. *Let L/K be normal. Then, any $\alpha \in L$ is a root of a separable $f \in K[T]$ that splits completely in L.*

Proof. Let $\alpha \in L$, $G := \mathrm{Aut}(L/K)$ and $\mathcal{O}(\alpha) := \{\sigma(\alpha) : \sigma \in G\}$. Additionally, let $f \in K[T]$ be the minimal polynomial of α.
Claim #1. $\mathcal{O}(\alpha)$ are roots of f.
Proof of Claim #1. We have $f(\alpha) = 0$, thus $\sigma(f(\alpha)) = 0$ since $\sigma \in G$ fixes 0. But $\sigma(f(\alpha)) = \sigma(\sum_i f_i\alpha^i) = \sum_i \sigma(f_i)\sigma(\alpha)^i = \sum_i f_i\sigma(\alpha)^i = f(\sigma(\alpha)) = 0$. Thus, $\sigma(\alpha)$ is also a root of f.
Claim #2. $f(T) = \prod_{\beta\in\mathcal{O}(\alpha)}(T - \beta)$.
Proof of Claim #2. Let $g(T) = \prod_{\tau\in G}(T - \tau(\alpha))$. Take any $\sigma \in G$. Then,

$$\sigma(g) = \sigma\prod_{\tau\in G}(T - \tau(\alpha)) = \prod_{\tau\in G}(T - \sigma\tau(\alpha)),$$

but clearly $\tau \mapsto \sigma\tau$ is a bijection, so $\forall_{g\in G}\ \sigma(g) = g$. Thus, $g \in K[T]$. Now, since all roots of g are roots of f, $g \mid f$. But since f is minimal and $g(\alpha) = 0$, $f \mid g$. Thus, $f = g$.

Theorem 1.6. *L/K is normal iff L is a splitting field of separable $f \in K[T]$.*

Proof. Suppose L/K is normal, then $[L : K] = n < \infty$, so $L = K(\alpha_1, \ldots, \alpha_n)$. Let $f_i \in K[T]$ be the minimal polynomial of α_i. Then, L is a splitting field of $f := \prod_{i=1}^n f_i$.
Indeed, $f \in K[T]$ since each $f_i \in K[T]$. It clearly splits in L since each f_i does. So we are left to show that f is separable. Suppose otherwise that there is a repeated root: $f(\beta) = f'(\beta) = 0$ for some $\beta \in L$. Then, β is a root of only one f_ℓ: if it were the root of f_i and f_j for $i \neq j$, f_i and f_j would be the minimal polynomial of β, meaning $f_i = f_j$. Thus,

$$f'(\beta) = \sum_{i=1}^n f_i'(\beta)\prod_{j\neq i} f_j(\beta) = f_\ell'(\beta)\prod_{i\neq\ell} f_i(\beta)$$

Since $f_\ell'(\beta) \neq 0, \prod_{i\neq\ell} f_i(\beta) = 0$, meaning some other $f_i(\beta) = 0$ for $i \neq \ell$. Contradiction.

Now suppose L is the splitting field of a separable $f \in K[T]$. Suppose n roots lie in L but not in K. We prove this fact by induction in n. If $n = 0$, the statement is obvious as $L = K$. Otherwise, assume $f = \prod_i f_i$ be the decomposition of f into irreducible polynomials in $K[T]$. Let $s := \deg f_1$. Without loss of generality, assume $s > 1$. Then, $[K(\alpha) : K] = s$. Now consider the extension $K \subseteq K(\alpha_1) \subseteq L$. Note that f_1 splits completely over $K(\alpha_1)$, so the polynomial f has strictly fewer roots outside the base field than over K. Thus, $L/K(\alpha_1)$ is normal. Thus, $\mathrm{Fix}(L/K(\alpha_1)) = K(\alpha_1)$.

Now, let $\alpha_1, \ldots, \alpha_s$ be roots of f_1. Then, by Theorem 8, there exist isomorphisms $\sigma_i : K(\alpha_1) \to K(\alpha_i)$, which in turn can be extended to automorphisms $\sigma_i : L \to L$.

Take any $\theta \in \mathrm{Fix}(L/K) \subseteq \mathrm{Fix}(L/K(\alpha_1)) = K(\alpha_1)$. Thus, $\theta = c_0 + c_1\alpha_1 + \cdots + c_{s-1}\alpha_1^{s-1}$. Apply σ_i to θ and use $\sigma_i(\theta) = \theta$.

1.3 Meeting #3 Notes

These notes are based on "Fields and Galois Theory" by J.S. Milne [Mil22].

Definition 1.3. *A **primitive n^{th} root of unity** in K is $\zeta_n \in K$ such that $ord(\zeta_n) = n$ in $K^\times$.*

Proposition 1.1. *Let K be a field of characteristic not dividing n and let L be the splitting field of $T^n - 1$. Then,*

(a) there exists a primitive nth root of unity in L;

(b) if ζ_n is a primitive nth root of unity in L, then $L = K(\zeta_n)$;

(c) L/K is a Galois extension and $Gal(L/K) \to (\mathbb{Z}/n\mathbb{Z})^\times$ is an injective homomorphism.

Proof. (a) The roots of $T^n - 1$ are distinct, because its derivative nT^{n-1} is zero only at $T = 0$ unless $\mathrm{char}(K) \mid n$, thus L contains n distinct nth roots of unity. They form a finite subgroup of $L^\times$, and thus they form a cyclic group. Generators are sought elements.

(b) The roots of $T^n - 1$ are the powers of ζ_n (in fact, ζ_n is the generator of a cyclic subgroup formed by roots of $T^n - 1$), thus $K(\zeta_n)$ contains them all.

(c) Since L is a splitting field of a separable polynomial $T^n - 1$, extension L/K is Galois. Now let $\sigma \in \mathrm{Gal}(L/K)$ and ζ_n be some primitive root. Then, $\sigma(\zeta_n) = \zeta_n^{i_\sigma}$ for some i_σ with $\gcd(i_\sigma, n) = 1$. Thus, we define the map $\mathrm{Gal}(L/K) \to (\mathbb{Z}/n\mathbb{Z})^\times$ by sending $\sigma(\zeta_n) = \zeta_n^{i_\sigma} \mapsto i_\sigma$. It is clearly a homomorphism.

1.4 Normal Basis

These notes are based on [Con08]. Let L/K be a Galois extension of degree n and $G := \mathrm{Gal}(L/K) = \{\sigma_1, \ldots, \sigma_n\}$. A **normal basis** for L/K is a basis consisting of a set of K-conjugate elements $G\beta := \{\sigma(\beta) : \sigma \in G\}$.

Example. *Consider $\mathbb{C}/\mathbb{R}$. The basis $\{1, i\}$ is **not** a normal basis since i is not the conjugate of 1, while $\{i, -i\}$ is a set of $\mathbb{R}$-conjugate elements that is not normal since they are not linearly independent. An actual example of the normal basis is $\{1+i, 1-i\}$.*

Example. *For prime p, the basis $\{1, \zeta_p, \ldots, \zeta_p^{p-2}\}$ is **not** a normal basis of $\mathbb{Q}(\zeta_p)/\mathbb{Q}$, but $\{\zeta_p, \zeta_p^2, \ldots, \zeta_p^{p-1}\}$ is a valid normal basis[a].*

[a] In general, the primitive roots of unity in the nth cyclotomic field form a normal basis iff n is square-free. For instance, for $\mathbb{Q}(\zeta_9)/\mathbb{Q}$, the set $\{\zeta_9, \zeta_9^2, \ldots, \zeta_9^8\}$ is not normal since $1 + \zeta_9^3 + \zeta_9^6 = 0$.

Example. *For $\mathbb{Q}(\sqrt{2}+\sqrt{3})/\mathbb{Q}$, the four $\mathbb{Q}$-conjugates of $\sqrt{2}+\sqrt{3}$ are not a normal basis since they add up to 0. An example of the normal basis is:*

$$\{1+\sqrt{2}+\sqrt{3}+\sqrt{6}, 1-\sqrt{2}+\sqrt{3}-\sqrt{6}, 1+\sqrt{2}-\sqrt{3}-\sqrt{6}, 1-\sqrt{2}-\sqrt{3}+\sqrt{6}\}.$$

The main theorem that we will prove is that *any finite Galois extension admits the normal basis*. But before that, we will need the following auxilary lemma.

Lemma 1.3. *Let L/K be a Galois extension with $G = Gal(L/K)$ and let $\{\omega_1, \ldots, \omega_n\}$ be the basis of L/K. Then, the n-tuples $(\sigma_1(\omega_i), \sigma_2(\omega_i), \ldots, \sigma_n(\omega_i))$ (for $i \in [n]$) are an L-basis for L^n (or speaking more algebraically, $\det(\sigma_i(\omega_j))_{ij} \neq 0$).*

Proof. Suppose these elements span $W \subseteq L^n$. We want to show that $W = L^n$, but suppose otherwise $W \subsetneq L^n$. In such case there is a non-zero functional $\varphi \in (L^n)^\vee$ that vanishes on W. Suppose $\varphi(\mathbf{v}) = (c_1, \ldots, c_n) \cdot \mathbf{v}$. If φ vanishes on W, then $(c_1, \ldots, c_n) \cdot (\sigma_1(\omega_i), \ldots, \sigma_n(\omega_i)) = 0$ for every $i \in [n]$. Thus, we get $\sum_{j=1}^n c_j \sigma_j(\omega_i) = 0$ for all i and by taking K-linear combinations, $\sum_{j=1}^n c_j \sigma_j(x) = 0$ for all $x \in L$. Due to the linear independence of characters, each c_j is zero and thus $\varphi = 0$.

Theorem 1.7. *Each finite Galois extension L/K admits the normal basis.*

Proof. Denote $G = \mathrm{Gal}(L/K) = \{\sigma_1, \ldots, \sigma_n\}$ and assume $\sigma_1 = \mathrm{id}_L$. The proof is split into two cases:

1. field K is infinite;
2. field K is finite (that is, $L \cong \mathbb{F}_{q^d}$ and $K \cong \mathbb{F}_q$).

Case #1 – Infinite field. Fix some $\beta \in L$. Let us explore what a K-linear dependence

relation would look like for fixed $\beta \in L$:

$$\sum_{j=1}^{n} a_j \sigma_j(\beta) = 0.$$

for some $\{a_j\}_{j\in[n]} \subseteq K$. Applying σ_i^{-1} to both sides gives new linear relations

$$\sum_{j=1}^{n} a_j \cdot \sigma_i^{-1}\sigma_j(\beta) = 0, \quad i \in [n].$$

Thus, collecting these together for each i yields $D(\beta)\mathbf{a} = 0$ for $D(\beta) = \{\sigma_i^{-1}\sigma_j(\beta)\}_{ij}$. For $G\beta$ to be a basis, we want for this equation to have the only solution $\mathbf{a} = 0$. In other words, we need to find some $\beta \in L$ such that $\det D(\beta) \neq 0$.

Write $\beta = \sum_{k\in[n]} b_k\omega_k$ where $\{\omega_k\}_{k\in[n]}$ is a basis of L/K. Consider the polynomial

$$\Delta(T_1, \ldots, T_n) := \det\left(\sum_{k=1}^{n} \sigma_i^{-1}\sigma_j(\omega_k) \cdot T_k\right) \in L[T_1, \ldots, T_n].$$

Since n-tuples $\sigma_i(\omega_j)$ form a basis for L^n, there exist some $d_1, \ldots, d_n \in L$ such that

$$\sum_{k=1}^{n} d_k(\sigma_1(\omega_k), \ldots, \sigma_n(\omega_k)) = (1, 0, \ldots, 0).$$

Or, component-wise, $\sum_{k=1}^{n} d_k\omega_k = 1$ but $\sum_{k=1}^{n} d_k\sigma(\omega_k) = 0$ for $\sigma \neq \mathrm{id}_L$. Thus:

$$\sum_{k=1}^{n} d_k \cdot \sigma_i^{-1}\sigma_j(\omega_k) = \delta_{ij}$$

and so $\Delta(d_1, \ldots, d_n) = \det \mathbb{I}_n = 1$. Thus, Δ is a non-zero polynomial. Consequently, since K is infinite, there should exist some $b_1, \ldots, b_n \in K$ such that $\Delta(b_1, \ldots, b_n) \neq 0$. Then,

$$0 \neq \Delta(b_1, \ldots, b_n) = \det\left(\sum_{k=1}^{n} b_k \cdot \sigma_i^{-1}\sigma_j(\omega_k)\right) = \det\left(\sigma_i^{-1}\sigma_j \sum_{k=1}^{n} b_k\omega_k\right) = \det\Big(\sigma_i^{-1}\sigma_j(\beta)\Big).$$

Thus, $G\beta$ is a normal basis for L/K.

Case #2 – Finite field. Recall that any finite Galois extension of finite fields is cyclic: say, $G = \langle\sigma\rangle$ (more concretely, σ is the Frobenius endomorphism $x \mapsto x^q$). We treat L as a $K[T]$-module by letting T act on L as σ, so $f(T)\alpha = f(\sigma)\alpha$. The polynomial $T^n - 1$ annihilates all of L since $\sigma^n(\alpha) = \alpha$ for every $\alpha \in L$. Note that there are no other lower-degree polynomials satisfying this property: if say $c_0 + c_1T + \cdots + c_mT^m$ also annihilates the whole L for $m < n$, then $\sum_{i=1}^{m} c_i\sigma^i(\alpha) = 0$, and so by linear independence each $c_i = 0$.

The annihilator ideal $\mathrm{Ann}_{K[T]}(L) = \{f \in K[T] : f(\sigma) \equiv 0 \text{ on } L\}$ is principal and due to observations above, $\mathrm{Ann}_{K[T]}(L) = (T^n - 1)$. From the commutative algebra (namely, from the theorem of finitely generated modules over PID), there exists some $\beta \in L$ such that $\mathrm{Ann}_{K[T]}(\beta) = (T^n - 1)$.

Consider the map $K[T] \to L, f \mapsto f(\sigma)\beta$. Its kernel is precisely $(T^n - 1)$, thus it induces the K-linear embedding $K[T]/(T^n - 1) \hookrightarrow L$. Moreover, as both sets are of dimension n (viewed as K-vector spaces), this map is also surjective. Thus, the basis of $K[T]/(T^n - 1)$ (which is $1, T, \ldots, T^{n-1}$) is mapped to the basis of L (which is now $\beta, \sigma(\beta), \ldots, \sigma^{n-1}(\beta)$), concluding the proof.

It was shown as the consequence on the meeting that

Theorem 1.8 (Hilbert's Theorem 90). $H^1(Gal(L/K), L^\times) = 0$.

1.5 Kummer Extensions

Further denote by $\mu_n(K) = \{x \in \overline{K} : x^n = 1\}$ the group of nth roots of unit of field K. Denote the primitive root of unity by ζ. We define the Kummer extensions as follows.

Definition 1.4. *Let n be a positive integer and let K be a field such that:*

1. *$char(K)$ does not divide n;*
2. *$\mu_n(K) \subseteq K^\times$.*

*An extension L/K is called a **Kummer extension** if for some set $\Delta \subseteq K^\times$,*

$$L = K(\sqrt[n]{\Delta}) \triangleq K(\sqrt[n]{d} : d \in \Delta).$$

First we consider Kummer extension by adjoining a single element $a \in K$ such that $T^n = a$ has a root in L (so that $\Delta = \{a\}$).

Lemma 1.4. *Consider the Kummer extension $L = K(\sqrt[n]{a})$ for some $a \in K^\times$. Then,*

(i) L/K is a Galois extension;

(ii) if $\alpha \in L$ is a root of $T^n = a$, then $\sigma \mapsto \sigma(\alpha)/\alpha$ induces embedding $Gal(L/K) \hookrightarrow \mu_n(K)$;

(iii) if a is of order n in $K^\times/(K^\times)^n$, then $Gal(L/K) = \langle\sigma\rangle$ is cyclic for $\sigma(\alpha) = \zeta\alpha$.

Remark 1.1. *Note that the point (i) is not obvious. Assume we remove the condition that K contains $\mu_n(K)$. Then, $K := \mathbb{Q}$ and $L = \mathbb{Q}(\sqrt[3]{2})$ do not induce the Galois extension L/K. This is fixed by letting $K := \mathbb{Q}(\zeta_3)$, then $Gal(\mathbb{Q}(\zeta_3, \sqrt[3]{2})/\mathbb{Q}(\zeta_3)) \cong \mathbb{Z}/3\mathbb{Z}$.*

Proof. *Part (i).* Since $\mu_n(K) \subseteq K$, the roots of $T^n - a$ are precisely $\{\alpha, \zeta\alpha, \ldots, \zeta^{n-1}\alpha\}$, which all trivially lie in L. Therefore, L is a splitting field of $T^n - a$. Clearly $T^n - a$ is separable since it has a derivative nT^{n-1}, which has the only root 0 whenever $\text{char}(n)$ does not divide n.

Part (ii). Denote $\phi : \text{Gal}(L/K) \to K^\times, \sigma \mapsto \sigma(\alpha)/\alpha$. Note that $\sigma(\alpha^n) = \sigma(a) = a$, meaning $\sigma(\alpha)^n = a$. Thus, $\sigma(\alpha)$ is a root of $T^n - 1$, so $\sigma(\alpha) = \zeta^i\alpha$ for some i. Thus, $\sigma(\alpha)/\alpha = \zeta^i \in \mu_n(K) \subseteq K^\times$.

To check injectivity, note that if $\phi(\sigma) = \text{id}_L$, then $\sigma(\alpha) = \alpha$. Since $L = K(\alpha)$, an automorphism that fixes the generator must be identity. Thus, $\ker(\phi) = \{\text{id}_L\}$.

Part (iii). It is fairly easy to see that $T^n - a$ is irreducible if a is of order n in $K^\times/(K^\times)^n$. Then, $\#\text{Gal}(L/K) = n$ and since $\text{Gal}(L/K)$ is isomorphic to a subgroup of $\mu_n(K)$, it must be isomorphic to the whole $\mu_n(K)$. Thus, $\text{Gal}(L/K) \cong \mu_n(K)$, and $\mu_n(K)$ is cyclic. There must be some σ such that $\phi(\sigma) = \zeta$, meaning $\sigma(\alpha) = \zeta\alpha$.

Thus, we have observed that if the field contains some appropriate element a of order n in $K^\times/(K^\times)^n$, then the corresponding Kummer extension is cyclic. Turns out the converse is also true.

Lemma 1.5. *Let L/K be a Galois extension with $Gal(L/K) \cong \mathbb{Z}/n\mathbb{Z}$ and K satisfies (i) and (ii) of Definition 1.4. Then, $L = K(\sqrt[n]{b})$ for some $b \in K$.*

Proof. We have already proven that every cyclic extension admits the normal basis. Therefore, let $L = K(\gamma)$ where $\{\sigma(\gamma) : \sigma \in \mathrm{Gal}(L/K)\}$ is a basis of L/K. Consider

$$\beta = \sum_{s=0}^{n-1} \zeta^s \sigma^s(\gamma).$$

Notice that $\sigma(\beta) = \zeta^{-1}\beta$. Moreover, $\beta \neq 0$ since $\{\gamma, \ldots, \sigma^{n-1}(\gamma)\}$ are linearly independent over K. Therefore, $\beta^n \in K$ and $\beta^r \notin K$ for $0 < r < n$. Thus $\beta^n = b$ is of order n in $K^\times/(K^\times)^n$; by Lemma 1, $K(\sqrt[n]{b})$ is a cyclic extension of degree n contained in L, so $L = K(\sqrt[n]{b})$.

Lemma 1.6. *$K(\sqrt[n]{a}) \cong K(\sqrt[n]{b})$ iff $a = b^r c^n$ for some $c \in K$ and $(r, n) = 1$.*

Proof. The "if" part is obvious, so we prove the "only if" part.

Suppose that $K(\alpha) = K(\beta)$ with $\alpha^n = a$ and $\beta^n = b$. Let σ be the generator of the Galois group with $\sigma\alpha = \zeta\alpha$, then $\sigma\beta = \zeta^i\beta$ for some i. Suppose $\beta = \sum_{j=0}^{n-1} c_j\alpha^j$, then $\sigma\beta = \sum_{j=0}^{n-1} c_j\zeta^j\alpha^j$, so we can only have $\beta = c_i\alpha^i$.

Lemma 1.7. *Let L/K be a finite extension with $G := Gal(L/K)$ abelian, of exponent dividing n. Let $G = \bigoplus_{i=1}^r G_i$. For each i, let $L_i = Fix(H_i)$ for $H_i := \bigoplus_{j\neq i} G_j$. Then,*

(i) $Gal(L_i/K) \cong G_i$;

(ii) $L_i = K(\alpha_i)$ with $\alpha_i^n = a_i \in K$;

(iii) $L = K(\sqrt[n]{\Delta})$ is a Kummer extension with $\Delta = \{a_1, \ldots, a_r\}$.

***Slogan:** If the base field contains roots of unity, any abelian extension of exponent dividing n is formed by extraction of roots of elements of K.*

Proof.

Part (i). $\mathrm{Gal}(L_i/K) \cong G/H_i \cong G_i$. First congruence is due to the fundamental theorem of Galois theory, the second simply by cancelling out everything besides G_i.

Part (ii). Let $n_i = \#G_i$. Since $n_i \mid n$ and K contains nth roots of unity, K also contains n_ith roots of unity. Therefore, by Lemma 1.5, $L_i = K(\sqrt[n_i]{b_i})$ for some $b_i \in K$.

Theorem 1.9. *There is a one-to-one correspondence*

$$\{\text{Abelian extensions of } K \text{ of exponent } n\} \leftrightarrow \{\text{Subgroups of } K^\times/(K^\times)^n\}.$$

Proof. Define two maps:

1. ψ that maps subgroups to abelian extensions: $\Delta \mapsto K(\sqrt[n]{\Delta})$;

2. φ that maps abelian extensions to subgroups: $L/K \mapsto (L^\times)^n \cap K^\times/(K^\times)^n$.

We show that both $\varphi \circ \psi$ and $\psi \circ \varphi$ are identity maps.

$\varphi \circ \psi = \mathrm{id}$. We show that $\varphi(\psi(\Delta)) = \Delta$. This is equivalent to showing $\varphi(L) = \Delta$ for $L = K(\sqrt[n]{\Delta})$. Clearly, since φ sends L to $(L^\times)^n \cap K^\times/(K^\times)^n$, we have $\Delta \subseteq \varphi(L)$. Now take any $b \in \varphi(L) \setminus \Delta$. Then, b induces an extension $K(\sqrt[n]{b}) \subseteq L$. By the isomorphism of Kummer extensions, if $K(\sqrt[n]{b}) \subseteq K(\sqrt[n]{\Delta})$, the class b must be generated by the classes of Δ. Thus, $b \in \Delta$, and so $\varphi(\psi(\Delta)) = \Delta$.

$\psi \circ \varphi = \mathrm{id}$. Let L/K be an abelian extension of exponent n. Then, $L = K(\sqrt[n]{\Delta})$ by Lemma 1.7 and then $\psi(K(\sqrt[n]{\Delta})) = \Delta$ (by the same argument as above) and thus $\varphi(\Delta) = K(\sqrt[n]{\Delta}) = L$.

1.5.1 Cohomologies

Lemma 1.8. $H^1(G, \mu_n(K)) \cong Hom(G, \mu_n(K)) \cong K^\times/(K^\times)^n$.

Proof. We apply this lemma by letting $L := \overline{K}$. There is an exact sequence

$$0 \to \mu_n(K) \to L^\times \xrightarrow{\nu : x \mapsto x^n} L^\times \to 0$$

Taking the cohomology functor $H(G, \cdot)$, we get

$$H^0(G, \mu_n(K)) \to H^0(G, L^\times) \xrightarrow{\nu} H^0(G, L^\times) \to H^1(G, \mu_n(K)) \to H^1(G, L^\times) \to \dots$$

Notice that G acts trivially on $\mu_n(K)$, $H^1(G, \mu_n(K)) \cong \mathrm{Hom}(G, \mu_n(K))$, so we get

$$\mu_n(K) \to K^\times \xrightarrow{\nu} K^\times \to \mathrm{Hom}(G, \mu_n(K)) \to 0.$$

That said, $\mathrm{Hom}(G, \mu_n(K)) \cong K^\times/(K^\times)^n$.

Chapter 2

Algebraic Power Series

2.1 Eisenstein's Theorem

Definition 2.1. *For field κ, we say that $f \in \kappa[\![x]\!]$ is **algebraic function over** κ iff f is algebraic over $\kappa(x)$ (rational functions over κ). That is, f satisfies $P(x, f(x)) = 0$ for $P \in \kappa[x]$.*

Example. *$x^{1/2}$ is not an algebraic function since it does not belong to $\kappa[\![x]\!]$. In turn, $f(x) := (1-4x)^{-1/2}$ is algebraic function over any field: first, $f \in \kappa[\![x]\!]$:*

$$(1-4x)^{-1/2} = \sum_{n \geq 0} \binom{2n}{n} x^n,$$

and it satisfies $(1-4x)f(x)^2 = 0$.

Example. *Assume κ has character p. Then, the function $f(x) = \sum_{n \geq 0} x^{p^n}$ is algebraic. Indeed, notice that $f(x) = x + f(x^p) = x + f(x)^p$, so f satisfies $f(x)^p - f(x) + x = 0$.*

Our goal is to show the following fact.

Theorem 2.1 (Eisenstein's Theorem)**.** *Let $f \in \mathbb{Q}[\![x]\!]$ be an algebraic power series. Then, there exist $N, M \in \mathbb{N}$ such that $Mf(Nx) \in \mathbb{Z}[\![x]\!]$.*

Without loss of generality, further assume $f(0) = 0$: otherwise, $f(x) = \frac{p}{q} + f^*(x)$ for $f^*(0) = 0$, and so if we prove that there exist $M, N \in \mathbb{N}$ such that $Mf^*(Nx) \in \mathbb{Z}[\![x]\!]$, then $(qM)f(Nx) = pM + q \cdot Mf^*(Nx)$ trivially lies in $\mathbb{Z}[\![x]\!]$.

2.1.1 Diagonals

Definition 2.2. *Let $f(x_1, \ldots, x_n) = \sum_{\vec{u} \in \mathbb{N}^n_{\geq 0}} a_{\vec{u}} \mathbf{x}^{\vec{u}} \in \mathbb{Q}[\![x_1, \ldots, x_n]\!]$ where by $\mathbf{x}^{\vec{u}}$ we denote $x_1^{u_1} \ldots x_n^{u_n}$. Define the **diagonal** of f as $\mathscr{D}\{f\}(t) := \sum_{m \geq 0} a_{m,\ldots,m} t^m$.*

Definition 2.3. *We say that $f \in \kappa[\![x]\!]$ is **diagonal of a rational function (DRF)** iff $f = \mathscr{D}\{F\}$ for some $F \in \mathbb{Q}[x, y]$.*

Our goal is to show that any algebraic $f \in \mathbb{Q}[\![x]\!]$ is a DRF of a specific rational function. Specifically, we show the following proposition.

Proposition 2.1. *Let $P \in \kappa[x, y]$ and let $\varphi(x) = \sum_{m \geq 1} c_m x^m \in \kappa[\![x]\!]$ satisfy $P(x, \varphi(x)) = 0$. If $P'_y(0, 0) \neq 0$, then*

$$\varphi = \mathscr{D}\left\{ y^2 P'_y(xy, y) / P(xy, y) \right\}.$$

Field κ above is arbitrary.

Proof. Write $P(x, y) = (y - \varphi(x))Q(x, y)$ with $Q \in \kappa[\![x]\!][y]$. We have $Q(0, 0) \neq 0$ since $\varphi(0) = 0$. Then,

$$\frac{1}{P} \cdot \frac{\partial P}{\partial y} = \frac{1}{y - \varphi(x)} + \frac{1}{Q} \cdot \frac{\partial Q}{\partial y}.$$

Substitute $x \mapsto xy$ and multiply everything by y^2:

$$\frac{y^2 P'_y(xy, y)}{P(xy, y)} = \frac{y^2}{y - \varphi(xy)} + \frac{y^2 Q'_y(xy, y)}{Q(xy, y)}.$$

Apply $\mathscr{D}\{\cdot\}$ to both sides. On the left we have the desired expression from the proposition statement. On the right:

$$\mathscr{D}\left\{ \frac{y^2}{y - \varphi(xy)} \right\} + \mathscr{D}\left\{ \frac{y^2 Q'_y(xy, y)}{Q(xy, y)} \right\}.$$

We show that the left diagonal is exactly φ while the right one is zero. Note:

$$\mathscr{D}\left\{ \frac{y^2}{y - \varphi(xy)} \right\} = \mathscr{D}\{y(1 - y^{-1}\varphi(xy))^{-1}\} = \mathscr{D}\left\{ \sum_{n \geq 0} y^{-n+1} \varphi(xy)^n \right\}.$$

Now notice the following: $\varphi(xy)$ is simply the sum $\sum_k \alpha_k x^k y^k$. Therefore, the polynomial $\varphi(xy)^n$ is also of form $\sum_k \beta_k x^k y^k$. Hence each $y^{-n+1}\varphi(xy)^n$ is of form $\sum_k \beta_k x^k y^{k-n+1}$. Diagonal only includes those terms corresponding to k's such that $k = k - n + 1$, implying $n = 1$. Therefore, we obtain quite a straightforward result:

$$\mathscr{D}\left\{ \frac{y^2}{y - \varphi(xy)} \right\} = \mathscr{D}\{\varphi(xy)\} = \varphi.$$

Now the remaining term is easy to analyze. Notice that since $Q(0, 0) \neq 0$, we can expand $H(x, y) := Q'_y(x, y)/Q(x, y)$ into some series $\sum_{(i,j) \in \mathbb{N}^2_{\geq 0}} \alpha_{ij} x^i y^j$. Substituting $x \mapsto xy$ and multiplying by y^2 we obtain $y^2 H(xy, y) = \sum_{(i,j) \in \mathbb{N}^2_{\geq 0}} \alpha_{ij} x^i y^{i+j+2}$, which contains no diagonal terms. Thus,

$$\mathscr{D}\left\{ \frac{y^2 Q'_y(xy, y)}{Q(xy, y)} \right\} = 0,$$

and the proposition is proven.

Proposition 2.2 (Corollary). *Any algebraic function $\varphi \in \mathbb{Q}[\![x]\!]$ is a DRF*[a].

[a] Here, we need to argue why $P'_y(0,0) \neq 0$.

2.1.2 Proof of Eisenstein's Theorem

Now, we are in position to prove Eisenstein's Theorem.

Proof. We have established that $\varphi = \mathscr{D}\{q(x,y)/r(x,y)\}$ for $q, r \in \mathbb{Q}[x,y]$. Let $\varphi(t) = \sum_{n\geq 0} \varphi_n t^n$. We need to find integers N, M such that $M\varphi_n N^n \in \mathbb{Z}$ for all integers $n \geq 0$.

From the construction in Proposition 2.1, the denominator $r(0,0) \neq 0$. Without loss of generality, let $r(0,0) = 1$ (otherwise we can rescale both numerator and denominator). Write $r(x,y) = 1 - s(x,y)$ where $s(0,0) = 0$. Now note:

$$\frac{q(x,y)}{r(x,y)} = \frac{q(x,y)}{1-s(x,y)} = q(x,y)\sum_{k\geq 0} s(x,y)^k.$$

Now, we are going to eliminate polynomial denominators. For concreteness, let

$$q(x,y) = \sum_{i,j} \frac{\alpha_{ij}}{\beta_{ij}} x^i y^j, \quad s(x,y) = \sum_{i,j} \frac{\gamma_{ij}}{\delta_{ij}} x^i y^j.$$

Let $J := \operatorname{lcm}(\beta_{ij})$ and $M := \operatorname{lcm}(\delta_{ij})$ (since q and s are polynomials in $\mathbb{Q}$, this definition is fully valid). Now note:

$$\varphi_n = \text{“coefficient of } x^n y^n \text{ in } q(x,y)\sum_{k\geq 0} s(x,y)^k\text{”}.$$

Since $s(0,0) = 0$, each term in s is of degree at least 1. Consequently, each term in $s(x,y)^k$ is of degree at least k. Since the term $x^n y^n$ has a total degree $2n$, the only terms that can contribute to it are $\{s(x,y)^k\}_{k\leq 2n}$. The coefficient of $x^n y^n$ in $q(x,y)s(x,y)^k$ has a denominator that divides MJ^k. Since there is a finite number of terms that contribute to the total $x^n y^n$, the total denominator divides MJ^{2n}. Consequently, $M(J)^{2n}\varphi_n \in \mathbb{Z}$.

Therefore, by taking $N := J^2$, we obtain the desired result.

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